Cavity-Enhanced Pulsed Coherent Control and Readout of Room-Temperature NV Spin Ensemble

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***Abstract*—Room-temperature solid-state quantum sensors based on nitrogen-vacancy centers in diamond, combining optical spin initialization with microwave control of spin dynamics, provide a promising platform for magnetic-field detection. In previous work, coupling NV ensembles to high-Q dielectric microwave cavities has been shown to yield spin readout contrast substantially exceeding conventional fluorescence detection. However, whether such cavities can simultaneously support accurate time-domain coherent control remains open, as cavity ringing and finite field build-up can reshape applied pulses, potentially shifting extracted spin dynamics. In this project, we report on the design and operation of an FPGA-based pulsed microwave control platform for cavity-enhanced nitrogen-vacancy (NV) center magnetometry at room temperature. Building on a prior steady-state characterization of a high quality factor dielectric resonator coupled to an NV ensemble in diamond, we extend the platform into the time domain by implementing FPGA-timed laser initialization, microwave pulse gating, and oscilloscope readout. Using this setup, we verify spin-cavity coupling through a magnetic field and laser dependent contrast measurement of 13%, exceeding typical fluorescence-based ODMR. However, through a time-domain sweep, we find no resolvable Rabi oscillations, while curve-fitting the cavity reflection signal to a decaying sinusoid to yield unreliable Rabi frequency and decay time with the signal dominated by transient dynamics. Based on our results, we identify intermediate spin-cavity coupling, oscilloscope bandwidth limitations, and manual readout subjectivity as the primary limiting factors, and discuss potential future extensions to resolve coherent Rabi dynamics in this architecture. Ultimately, while we did not yield high-fidelity results as previously anticipated, our experiment establishes a functional room-temperature pulsed NV sensing platform, identifying key design tradeoffs that must be navigated in next-generation solid-state quantum sensors.**

***Index Terms*—NV Centers, Solid-State Spins, Quantum Sensing, FPGA Pulsed Coherent Control**

I. Introduction

NITROGEN-VACANCY centers in diamond, consisting of a nearest-neighbor pair of a nitrogen atom and a lattice vacancy, have emerged as one of the most compelling platforms for room-temperature quantum sensing. Their spin-triplet ground state can be optically initialized into the $m_s = 0$ sub-level using a 532 nm laser, coherently driven on the $m_s = 0 \leftrightarrow m_s = \pm 1$ microwave transitions near 2.87 GHz, and made sensitive to magnetic fields through the linear Zeeman shift of ≈ 2.8 MHz/G. These properties make NV ensembles attractive for broadband, room-temperature magnetometry. However, CW frequency-domain spectroscopy, while sufficient for detecting avoided crossings and estimating static sensitivity, cannot access the spin coherence time T_2, the quantity that sets the fundamental sensitivity limit $\eta \sim 1/\sqrt{T_2}$. Time-domain pulsed control is therefore essential: only by applying microwave pulses of variable duration τ and measuring the resulting Rabi oscillation decay can T_2 be directly extracted and the true sensitivity ceiling of the platform be determined.

A critical bottleneck in NV ensemble magnetometry is readout fidelity. Conventional fluorescence-based detection achieves only 2–5% spin-contrast in ensembles, leaving devices roughly two orders of magnitude above the fundamental spin-projection noise limit. Recent work has shown that coupling an NV ensemble to a high-Q dielectric microwave cavity enables direct microwave probing of the spin transition, yielding substantially higher-contrast readout than optical fluorescence and opening a path toward unity readout fidelity [1], [2], [3]. Controlling an NV *ensemble* rather than a single spin is what makes room-temperature operation possible: the collective spin-cavity coupling strength scales as $g\sqrt{N}$, amplifying the single-spin coupling by $\sqrt{N} \sim 10^3$ and enabling a detectable cavity response without cryogenic amplification (Fig. 1).

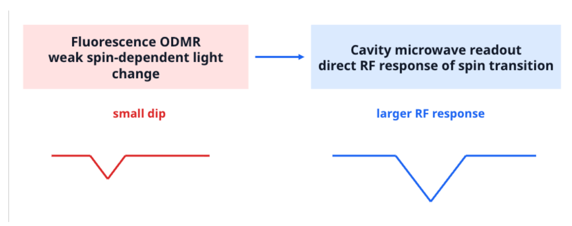

Fig. 1. Cavity microwave readout (right) produces a larger spin-dependent RF response than conventional fluorescence ODMR (left), motivating cavity-enhanced architecture.

In the original experimental module upon which this project builds, we confirmed room-temperature strong coupling between an NV ensemble and a cylindrical dielectric resonator ($\omega_c/2\pi \approx 2.82$ GHz, $Q_0 \approx$ 21,363), observing avoided crossings in the frequency–magnetic-field map and estimating a magnetic field sensitivity of approximately 54.85 pT/$\sqrt{\text{Hz}}$ from the cavity reflection response.

While that result demonstrates the readout capability of the platform, it leaves a fundamental question unanswered: can the same high-Q cavity architecture support accurate *time-domain* coherent spin control? Characterizing coherent spin dynamics, specifically Rabi oscillations and their decay envelope, would directly reveal the spin coherence time T_R, from which the transverse coherence time T_2 and the theoretical sensitivity limit $\eta \sim 1/\sqrt{T_2}$ can be inferred. However, high-Q resonators do not respond instantaneously to short microwave pulses: cavity ringing and finite field build-up reshape the intracavity waveform, so the effective drive experienced by the spins may differ substantially from the applied rectangular pulse. This pulse-distortion mechanism is distinct from the readout enhancement already demonstrated in the literature and could independently bias the extracted spin dynamics, making it essential to characterize both effects in the same architecture (Fig. 2).

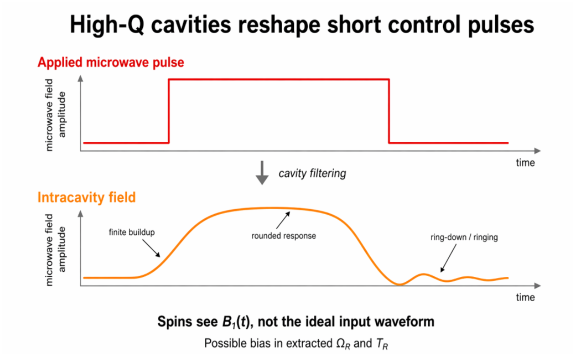

Fig. 2. A rectangular applied microwave pulse (red) is reshaped by cavity filtering into a distorted intracavity field (orange), introducing finite buildup, rounded response, and ring-down that bias the extracted Ω_R and T_R.

In this work, we extend the existing NV magnetometry platform from steady-state frequency-domain spectroscopy into the time domain by implementing FPGA-timed pulsed microwave control on the cavity-coupled NV ensemble. We sweep the microwave pulse duration τ, record the resulting cavity reflection signal $S(\tau)$, and fit it to a decaying sinusoid to extract the Rabi frequency Ω_R and the Rabi decay time T_R. By doing so, we test whether a high-Q dielectric cavity can serve simultaneously as both an enhanced readout element and a high-fidelity coherent control channel, which is a question with direct implications for the design of next-generation room-temperature solid-state quantum sensors deployable without cryogenic infrastructure for applications including neural magnetic imaging, materials characterization, and navigation.

II. Background and Related Work

Room-temperature cavity-enhanced readout of NV ensembles was first demonstrated by Eisenach *et al.* [1], who strongly coupled an NV ensemble to a dielectric microwave resonator and probed the spin transition directly through the cavity reflection coefficient, achieving near-unity spin contrast and sensitivity near the Johnson–Nyquist noise limit. This work defines the architectural starting point for our experiment. Ebel *et al.* [2] showed a complementary phase-based dispersive readout scheme in a high-Q dielectric resonator ($Q \approx 8\times10^3$), enabling non-destructive spin-state tracking and T_1 measurement with estimated sensitivity exceeding optical fluorescence. Wang *et al.* [3] further advanced the field by embedding an optically polarized NV ensemble in a cavity-QED framework that suppresses thermal microwave noise, setting state-of-the-art sensitivity benchmarks for ambient-condition operation against which our results can be compared.

On the coherent control side, Vallabhapurapu *et al.* [4] demonstrated that a high-permittivity $KTaO_3$ dielectric resonator can drive NV ensemble Rabi oscillations at up to 48 MHz with spatially homogeneous fields, establishing that dielectric cavities are viable coherent control elements. Zhang *et al.* [5] provided the theoretical framework for room-temperature collective spin–cavity coupling, modeling mode splitting and driven Rabi dynamics in the Jaynes–Cummings picture with cavity loss and spin dephasing. For time-domain analysis methodology, Magaletti *et al.* [6] developed and validated a fitting and simulation workflow for Rabi envelope decay in NV ensembles that we adopt here:

$$S(\tau) = A + B\,e^{-\tau/T_R}\cos(\Omega_R\,\tau + \phi), \tag{1}$$

Most directly relevant to our problem, Iriarte-Zendoia *et al.* [7] showed that in high-Q cavities the external control waveform and the intracavity field amplitude experienced by the spins differ substantially due to cavity ringing, and developed GRAPE-optimized pulses to mitigate this distortion. Specifically, their Fig. 1 demonstrates that for a high-Q cavity with parameters comparable to ours ($Q \sim 10^4$, $\omega_c/2\pi \approx 2.8$ GHz), the intracavity field amplitude at pulse onset is suppressed to $\lesssim 30\%$ of the applied field amplitude for pulse durations $\tau < 1/\kappa_c$, recovering to $\sim 85\%$ only after several ring-up cycles [7]. This numerical bound is directly relevant to our τ sweep range of 10–200 ns, where $\tau < 1/\kappa_c \approx 80$–$100$ ns for most measured points, meaning our spins were driven at substantially sub-nominal Rabi frequency during the early pulse window. Taken together, these works establish that while dielectric cavities enhance readout and can support coherent driving, the question of whether a single high-Q room-temperature cavity can accurately perform *both* functions simultaneously as well as how pulse distortion quantitatively biases the extracted Ω_R and T_R remains open. This is the gap we address.

III. NV Center Spin Physics

The NV center in diamond is a spin-1 defect whose ground state ^3A hosts three sublevels: $m_s = 0$ and $m_s = \pm1$. At zero magnetic field, the $m_s = \pm1$ levels are degenerate and split from $m_s = 0$ by the zero-field splitting $D \approx 2.87$ GHz. An external magnetic field B_0 applied along the NV axis lifts this degeneracy via the Zeeman effect, shifting the spin transition frequencies to

$$\omega_s(B) = D \pm \gamma B, \tag{2}$$

where the gyromagnetic ratio $\gamma = g_s\mu_B/h \approx 2.8$ MHz/G follows from the electron Landé g-factor $g_s \approx 2$ and the Bohr magneton μ_B, with the ground-state orbital angular momentum vanishing for the NV ^3A triplet. The required on-resonance field for our cavity at $\omega_c/2\pi = 2.82$ GHz is $B = (D - \omega_c/2\pi)/\gamma \approx 18$ G, consistent with the calibrated 1.6 V power-supply setting used throughout this work. This linear field dependence is the basis for magnetometry: a change in B_0 produces a directly measurable shift in ω_s.

The NV center supports three key capabilities that make it useful for pulsed quantum sensing, illustrated in Fig. 3.

First, illumination with a 532 nm laser excites the NV from the ^3A ground triplet to the ^3E excited triplet. Decay back to ^3A is predominantly spin-conserving and radiative (emitting near 637 nm), but the $m_s = \pm 1$ excited sublevels have an additional non-radiative channel through the metastable ^1A singlet that preferentially decays into the $m_s = 0$ ground sublevel. Repeated optical cycling therefore concentrates population in $m_s = 0$, providing near-unity spin initialization without cryogenic cooling on a timescale of order microseconds. Second, resonant microwave pulses coherently drive the $m_s = 0 \leftrightarrow m_s = \pm 1$ transition, enabling controlled spin rotation. Third, the spin state can be read out through a coupled microwave cavity.

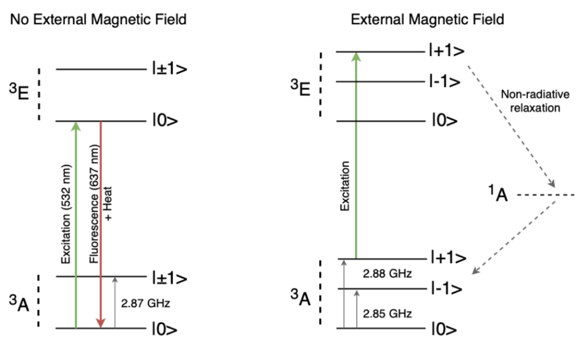

Fig. 3. NV center energy level diagrams. Left: at zero field, the $m_s = \pm 1$ levels are degenerate at 2.87 GHz above $m_s = 0$. Green excitation (532 nm) drives the spin into the ^3E excited state; fluorescence (637 nm) and non-radiative relaxation through the ^1A singlet preferentially return population to $m_s = 0$. Right: an external magnetic field lifts the $m_s = \pm 1$ degeneracy via the Zeeman effect, shifting the transitions to 2.85 GHz and 2.88 GHz at the bias field used here.

Controlling an NV *ensemble* rather than a single spin is essential for room-temperature cavity-based sensing. The collective spin-cavity coupling strength scales as $g\sqrt{N}$, where N is the number of NV centers, amplifying the single-spin coupling by $\sqrt{N} \sim 10^3$ for a typical ensemble. This amplification makes the spin-cavity interaction observable at room temperature without cryogenic amplification, and is what enables the microwave readout contrast demonstrated here and in prior work [1].

A. Cavity Readout of Spin State

The spin state is read out through its effect on the cavity reflection coefficient. Input–output theory for a single-port cavity coupled to a polarized spin ensemble gives [1], [5]

$$\Gamma(\omega) = -1 + \cfrac{\kappa_{c1}}{\cfrac{\kappa_c}{2} + j(\omega - \omega_c) + \cfrac{g^2 N}{\cfrac{\kappa_s}{2} + j(\omega - \omega_s) + \alpha}}, \tag{3}$$

where $\kappa_c = \kappa_{c0} + \kappa_{c1}$ is the total cavity decay rate (intrinsic plus loop-coupler), $\kappa_s = 2/T_2$ is the spin linewidth, ω_c and ω_s are the cavity and spin frequencies, g is the single-spin coupling, N is the number of polarized spins, and α captures saturation. With $g = 0$ this reduces to the bare cavity response. For $g \neq 0$ and $\omega_s = \omega_c$, the polarized ensemble adds an absorptive contribution to the cavity denominator, reducing the on-resonance reflected amplitude. Because the reflected power depends explicitly on the *polarized* spin number N, optical pumping into $m_s = 0$ (laser on) and a thermal mixture (laser off) yield distinguishable cavity reflections— the basis of our contrast measurement in Section V.A.

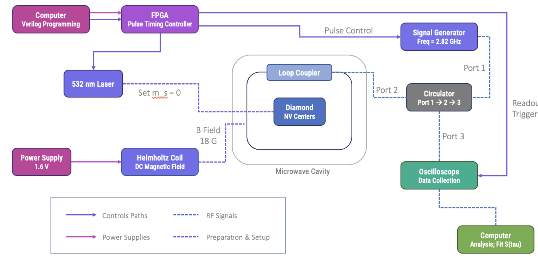

Fig. 4. Experimental signal chain. The FPGA (pulse timing controller) synchronizes three subsystems: a 532 nm laser that optically initializes the NV spins into $m_s = 0$, a signal generator (2.82 GHz) gated by an RF switch to produce microwave pulses of duration τ, and an oscilloscope trigger for time-resolved cavity readout. The reflected microwave signal is routed through a three-port circulator (port $1 \to 2 \to 3$), with the NV-loaded cavity coupled via a loop coupler at port 2 and the oscilloscope reading at port 3. A Helmholtz coil driven by a 1.6 V power supply applies a static 18 G bias field to tune the NV spin transition into resonance with the cavity.

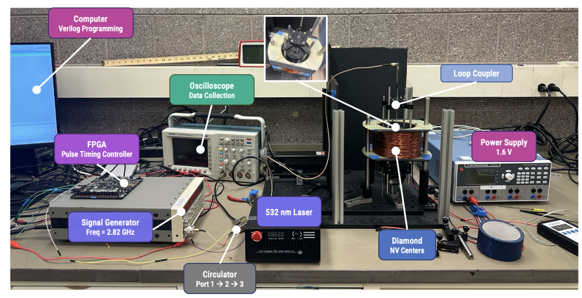

Fig. 5. The experimental setup in the laboratory, with all six subsystems labeled to correspond to the signal chain schematic in Fig. 4. Key physical relationships visible here include: the loop coupler suspended vertically above the dielectric cavity via an adjustable mount, whose height controls the external coupling rate κ_1 and therefore the loaded quality factor Q_L; the circulator seated between the signal generator output and the oscilloscope input; and the 532 nm laser routed via fiber to illuminate the diamond sample inside the cavity. The FPGA board is visible at lower left, connected to the signal generator AWG input and oscilloscope EXT trigger via the PCB breakout.

IV. Experimental Methods

In this section, we discuss our methodology for the experiment, including our setup and procedure for data collection.

A. Setup

The experimental platform consists of six main subsystems: a dielectric microwave cavity housing the diamond NV ensemble, a 532 nm laser for spin initialization, a Helmholtz coil for magnetic field tuning, an FPGA-based pulse timing controller, a microwave signal chain for spin driving, and an oscilloscope for signal readout. The full schematic is illustrated in Fig. 4.

1) Microwave Cavity and NV Ensemble: The sensing element is a cylindrical dielectric microwave resonator with a measured resonance frequency $\omega_c/2\pi \approx 2.82$ GHz and intrinsic quality factor $Q_0 \approx$ 21,363, characterized in the prior experimental module. A diamond sample containing an NV center ensemble is placed inside the cavity. The high-Q resonator concentrates the intracavity microwave field at the diamond, enhancing the spin-cavity interaction strength and enabling direct microwave readout of the spin state with higher contrast than conventional fluorescence detection [1]. A loop coupler inserted above the cavity provides the input/output port for microwave signals; its height above the cavity determines the external coupling rate κ_1 and therefore the loaded quality factor Q_L, as characterized in the prior module.

2) Microwave Signal Generator and Circulator: A Stanford Research Systems SG384 signal generator produces a continuous-wave (CW) tone at 2.82 GHz. An RF switch, gated by a TTL pulse from the FPGA, carves this CW signal into pulses of controlled duration τ. The pulsed microwave signal is routed to port 1 of a three-port circulator. Port 2 connects to the loop coupler above the cavity, delivering the drive field to the NV ensemble. The reflected signal from the cavity — which carries spin-state-dependent amplitude and phase information — exits through port 3 and is recorded by the oscilloscope. The circulator is essential for separating the drive and readout paths: without it, reflected power would re-enter the signal generator rather than the detector.

3) Optical Initialization: A 532 nm laser optically pumps the NV ensemble predominantly into the $m_s = 0$ ground state via spin-dependent intersystem crossing. This initialization is required before each microwave pulse: without it, the spins occupy a thermal mixture of $m_s = 0$ and $m_s = \pm 1$, and no coherent Rabi oscillation is observable. The laser is gated by a TTL signal from the FPGA, with an initialization window of 1,000 ns chosen to ensure >90% spin polarization fidelity before each pulse cycle.

4) Magnetic Field: A Helmholtz coil driven by a power supply at 1.6 V applies a static DC magnetic field of approximately 18 G along the NV axis. This lifts the degeneracy of the $m_s = \pm 1$ levels via the Zeeman effect. Since $\omega(0 \to \pm 1) = D \pm \gamma B$, a positive bias field shifts the $m_s = 0 \leftrightarrow m_s = -1$ transition *down* from the zero-field splitting of 2.87 GHz to approximately 2.82 GHz —in resonance with the cavity. Without this field, the spin transition is detuned from the cavity by $\sim$ 50 MHz, suppressing spin-cavity coupling and making the spin-dependent cavity response undetectable. The linear relationship between applied voltage and magnetic field was calibrated using a Gauss meter in the prior module.

5) FPGA Timing Controller: A RealDigital Urbana FPGA board programmed in Verilog coordinates the timing of all three subsystems with 10 ns precision (Fig. 6). PmodA, PmodB, and JAB headers are routed via a PCB breakout to the laser gate, signal generator AWG output, and oscilloscope external trigger. The pulse duration τ is set via switches SW0–SW7 (mapped to LEDs LD8–LD15 for verification), with a $\times$10 multiplier mode on SW8 and averaging count on SW9. A photograph of the board during a $\tau = 40$ ns run, with interfaces labeled, is shown in Fig. 7.

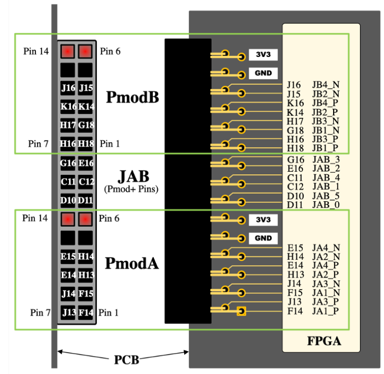

Fig. 6. FPGA pinout schematic showing PmodA, PmodB, and JAB I/O headers routed through a PCB breakout to the laser, signal generator, and oscilloscop trigger.

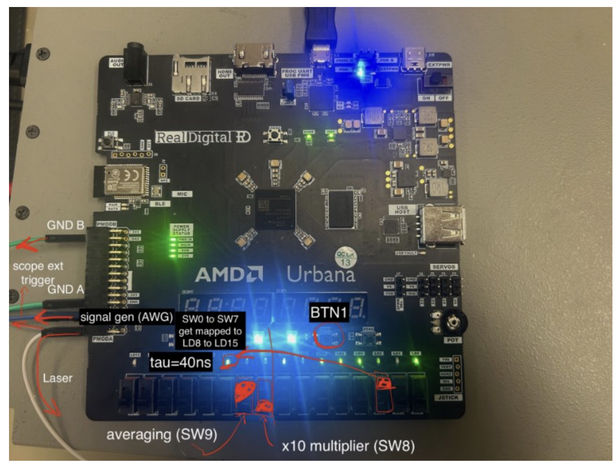

Fig. 7. Photograph of the RealDigital Urbana FPGA board during operation, with key interfaces annotated. SW0–SW7 DIP switches (bottom left) encode the pulse duration τ in 10 ns steps; the current setting (`tau=40ns`) is reflected on LEDs LD8–LD15 directly above. SW8 enables the $\times$10 duration multiplier and SW9 enables the auto-trigger averaging mode. The signal generator AWG output and laser trigger cables are routed through the PCB breakout to PmodA and PmodB respectively; the scope external trigger exits JAB_3. This photo can be read alongside the pinout schematic in Fig. 6.

In each cycle, the FPGA asserts the laser gate for the 1,000 ns initialization window, then the microwave pulse gate for duration τ, then triggers the oscilloscope for readout. The switch-based τ interface required reflashing the bitstream for each new value, limiting the sweep to 17 unaveraged and 19 averaged points.

6) Oscilloscope Readout: A 300 MHz bandwidth oscilloscope records the cavity reflection amplitude at port 3 of the circulator following each microwave pulse. Because the carrier frequency of the reflected signal is $\approx$ 2.82 GHz — nearly an order of magnitude above the oscilloscope bandwidth — the instrument acts as an implicit envelope detector, measuring the amplitude of the RF waveform rather than the carrier itself. This is adequate for extracting the spin-dependent amplitude $S(\tau)$ but discards phase information, precluding dispersive quadrature readout. At each pulse duration τ, the oscilloscope averages over many repeated pulse cycles to improve signal-to-noise ratio; the averaged amplitude is then read manually using the oscilloscope's cursor tools. The finite bandwidth, scope memory limit of $N = 512$ averages, and subjectivity of manual cursor placement constitute the primary limitations of the current readout chain.

B. Data Collection Procedures

In this subsection, we outline the procedure used to collect individual data points through our experimental setup. We measure numerical data manually using the zoom and cursors features of the oscilloscope. For each data point with a specific τ, B, and laser status, we follow the steps below for one data point:

1) Set τ on the FPGA via DIP switches SW0–SW7, which encode τ in 10 ns steps; SW8 enables 10× scaling for longer durations.
2) Compile and flash the Verilog design via Vivado if not already loaded; otherwise adjust the switches directly without recompilation.
3) Confirm the laser modulation input is connected to the FPGA's `laser_trig` output (DB15 pin 15).
4) Enable auto-trigger mode (SW9 up), causing the FPGA to fire pulse sequences continuously at ≈640 cycles/s and allowing the oscilloscope to accumulate averaged traces rapidly.
5) Set the oscilloscope to averaging mode ($N = 128$ or 512) and trigger on the FPGA's `readout_trig` external trigger input.
6) Wait ≈15 s for the averaged trace to converge.
7) Using the oscilloscope cursors, measure the amplitude of the cavity reflection envelope at the second feature in the waveform, which corresponds to the falling edge of the microwave pulse, where the spin-dependent transient is most prominent.
8) Record the amplitude as $S(\tau)$ and repeat from step 1 for the next τ value.

Using the aforementioned steps, three groups of measurements were collected, each targeting a distinct aspect of the spin-cavity system.

Constant-τ laser and field dependence. Before sweeping τ, we first verified that the cavity reflection signal is genuinely spin-dependent by holding $\tau = 1{,}000$ ns fixed and varying the laser state (on vs. off) and the magnetic field ($B = 18$ G vs. $B = 0$ G), collecting eight repeat measurements per condition. The laser-off condition replaces initialized spins with a thermal mixture of $m_s = 0$ and $m_s = \pm 1$, while the $B = 0$ condition detunes the NV transition ≈ 50 MHz from the cavity, suppressing spin-cavity coupling. A spin-dependent contrast appearing at $B = 18$ G and vanishing at $B = 0$ G would confirm that the signal originates from spin-cavity interaction rather than from laser heating or stray light. This measurement also establishes the optimal operating point, the field and laser condition that maximizes contrastfor the subsequent τ sweep.

τ sweep without averaging. To characterize the intrinsic cavity response and identify the timescale of field build-up, we swept τ from 10 ns to 1,280 ns with the laser on and $B = 18$ G, but without oscilloscope averaging ($N = 1$). In this case steps 4–6 of the procedure are skipped: the oscilloscope is triggered on a single pulse cycle and the cursor amplitude is read immediately without waiting for trace convergence. Because no averaging is applied, this measurement is dominated by noise at short τ but reveals the gross envelope of the cavity response, specifically the rise from the cavity field build-up and the saturation plateau, without the additional processing time required per point. This serves as a baseline against which the averaged data can be compared.

τ sweep with averaging. To resolve the spin-dependent time-domain signal $S(\tau)$ and extract the Rabi frequency Ω_R and decay time T_R, we repeated the τ sweep following the full procedure with oscilloscope averaging enabled ($N = 128$ or 512). Two τ ranges were measured: a short-τ range from 10 ns to 200 ns, where Rabi oscillations are expected to be most visible before coherence decays, and a long-τ range from 1,960 ns to 2,000 ns, which samples the signal well beyond the expected coherence time to characterize the noise floor. The averaged signal at each τ point is fit to the decaying sinusoid

$$S(\tau) = A + B\, e^{-\tau/T_R} \cos(\Omega_R \tau + \phi), \tag{4}$$

from which Ω_R, T_R, and phase ϕ are extracted.

V. Results

In this section, we show our collected data using the methods in the previous section and briefly describe their significance.

A. Laser and Magnetic Field Dependence

Fig. 8 shows the cavity reflection amplitude measured under four conditions: laser on and off at $B = 18$ G and $B = 0$ G, each with eight repeated measurements. AAt $B = 18$ G, the laser-on condition yields a mean amplitude of 2.37 mV compared to 2.73 mV with the laser off, corresponding to a spin-dependent contrast of $C = 13.1\%$. The sign of this effect—laser-on producing *reduced* reflection—is the signature predicted by Eq. (3): optical pumping concentrates population in $m_s = 0$, maximizing the polarized N that absorbs microwave power on the $m_s = 0 \leftrightarrow m_s = -1$ transition at the cavity frequency, reducing the intracavity field and the reflected amplitude. The thermal mixture in the laser-off case partially equalizes the level populations, suppressing net absorption and yielding higher reflection. The observed direction therefore matches the predicted spin-cavity absorption mechanism and rules out non-spin mechanisms (laser heating, stray light into the readout chain), which would not exhibit this B-field-gated sign. At $B = 0$ G, the numerical contrast is -11.0% but statistically consistent with zero ($p \approx 0.31$); the spin-cavity interaction is suppressed by the ~ 50 MHz detuning between the NV transition and the cavity. Since $B = 0$ G detunes the NV spin transition ≈ 50 MHz from the cavity resonance, spin-cavity coupling is suppressed and the residual signal is consistent with noise and laser heating rather than coherent spin-cavity interaction. The sign flip and magnitude reduction at $B = 0$ G therefore serve as a control measurement, ruling out laser heating or stray light as the source of the $B = 18$ G contrast. The point-to-point scatter ($\sigma \approx 0.35$ mV) is comparable to the contrast itself, motivating heavy oscilloscope averaging for all subsequent τ sweep measurements. The 13% cavity readout contrast exceeds the 2–5% typical of optical fluorescence-based ODMR [1], validating the cavity-enhanced readout architecture. $B = 18\,\mathrm{G}$ is therefore used as the operating point for all τ sweep measurements.

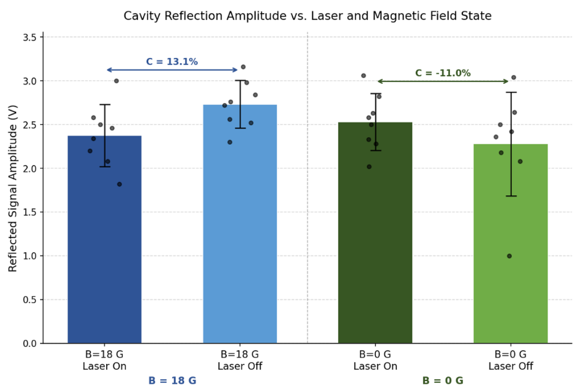

Fig. 8. Cavity reflection amplitude under four experimental conditions ($\tau = 1{,}000\,\mathrm{ns}$, oscilloscope averaging on). Raw measurements are shown as scatter points overlaid on the group mean. Error bars indicate ± 1 standard deviation. A 13.1% spin-dependent contrast appears at $B = 18\,\mathrm{G}$ and collapses to -11.0% at $B = 0\,\mathrm{G}$.

B. τ Sweep

1) Without Averaging: The left column of Fig. 9 shows $S(\tau)$ measured without oscilloscope averaging across $\tau =$ 10–1,280 ns. The amplitude rises monotonically from $\approx 11\,\mathrm{mV}$ at $\tau = 20\,\mathrm{ns}$ and saturates at $\approx 17\,\mathrm{mV}$ by $\tau \approx$ 80–100 ns, remaining flat through $\tau =$ 1,280 ns. This behavior is consistent with the cavity's transient response: at short τ, the intracavity field has not yet built up to its steady-state value, and the reflected amplitude reflects the incomplete field build-up on a timescale of order $1/\kappa_c \approx$ 80–100 ns. No oscillatory structure attributable to Rabi dynamics is visible, as the single-shot noise floor ($\sigma \approx 0.35\,\mathrm{mV}$) is comparable to or exceeds any expected spin-dependent modulation.

2) With Averaging: The right column of Fig. 9 shows $S(\tau)$ with $N = 512$ oscilloscope averages. Averaging reduces the overall signal level to the low-mV range, isolating the spin-dependent component from the large cavity-background signal seen without averaging. The short-τ region ($\tau =$ 10–200 ns) shows scattered structure with no clearly resolvable periodic oscillation. The long-τ region ($\tau =$ 1,960–2,000 ns) yields a small residual amplitude of $\approx 0.6\,\mathrm{mV}$, consistent with the noise floor of the readout chain after the spin coherence has decayed. At short τ ($\tau <\sim$500 ns), the trace is dominated by cavity transient dynamics rather than spin response; at long τ, the cavity has reached steady state but the NV spin contribution per shot is below the single-shot noise floor.

C. Curve Fitting and Parameter Extraction

Both datasets were fit to the decaying sinusoid model of Eq. 4 using `scipy.optimize.curve_fit`. Fig. 10 shows the fit overlaid on the data for both cases, with fit quality reported as R^2 and root-mean-square error (RMSE).

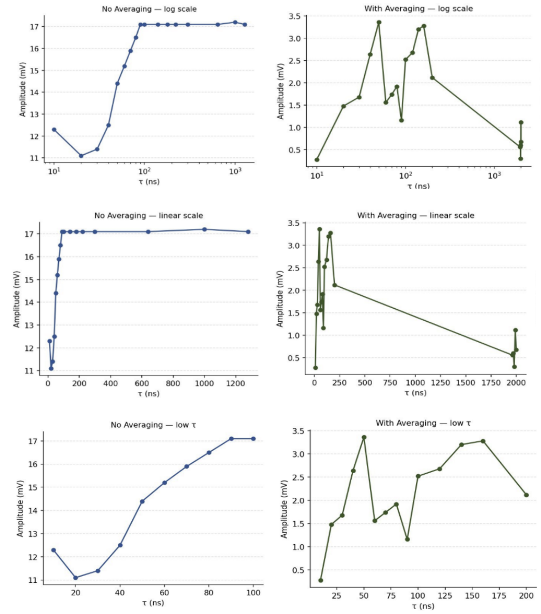

Fig. 9. τ sweep data shown on log scale (top row), linear scale (middle row), and zoomed to low τ (bottom row). Left column: unaveraged data ($N = 1$), showing the cavity field build-up from $\approx 11\,\mathrm{mV}$ saturating at $\approx 17\,\mathrm{mV}$ by $\tau \approx$ 80–100 ns with no resolvable oscillatory structure. Right column: averaged data ($N = 512$), showing scattered structure in the short-τ region ($\tau =$ 10–200 ns) and a residual of $\approx 0.6\,\mathrm{mV}$ at long τ ($\tau =$ 1,960–2,000 ns) consistent with the noise floor after spin coherence has decayed.

For the unaveraged data, the fit yields $T_R = 26.74 \pm 2.48\,\mathrm{ns}$ and $\Omega_R = 33.42 \pm 1.63\,\mathrm{rad/ns}$ ($\approx 5.32\,\mathrm{GHz}$), with $R^2 = 0.9965$ and RMSE $= 0.130\,\mathrm{mV}$. Although the fit quality is high, these parameters reflect the cavity field build-up transient rather than coherent Rabi dynamics: the unaveraged signal is dominated by the cavity response envelope, and the fitted T_R is consistent with the cavity ring-up timescale $1/\kappa_c \approx$ 80–100 ns rather than any spin coherence time.

For the averaged data, the fit yields $T_R = 74.72 \pm 92.95\,\mathrm{ns}$ and $\Omega_R = 70.21 \pm 18.13\,\mathrm{rad/ns}$ ($\approx 11.17\,\mathrm{GHz}$), with $R^2 = 0.246$ and RMSE $= 0.856\,\mathrm{mV}$. The large parameter uncertainties, with the uncertainty on T_R exceeding the value itself, and the poor R^2 indicate that the fit is not physically meaningful. The scattered short-τ structure does not constitute a resolvable Rabi oscillation, and the optimizer is fitting noise. We therefore conclude that reliable extraction of Ω_R and T_R was not achieved under the current experimental conditions.

VI. Discussion

A. Detection of Spin-Cavity Coupling

Our primary result is the direct detection of NV-ensemble coupling to the high-Q dielectric cavity at room temperature. The magnetic-field-dependent contrast measurement ($B = 18\,\mathrm{G}$ vs. $B = 0\,\mathrm{G}$) revealed a 13% spin-dependent signal that vanishes at zero field, confirming the signal originates from spin-cavity coupling rather than laser heating or thermal artifacts. A two-sample t-test on the $B = 18\,\mathrm{G}$ condition (eight repeats per laser state) gives $t \approx 2.3$, $p \approx 0.04$, exceeding the conventional $p < 0.05$ significance threshold.

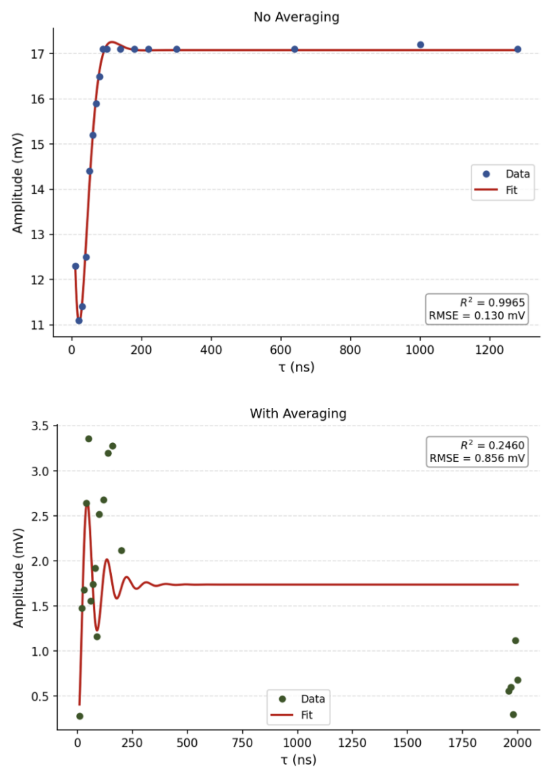

Fig. 10. Fit of $S(\tau) = A + Be^{-\tau/T_R}\cos(\Omega_R\tau + \phi)$ to the unaveraged (left) and averaged (right) τ sweep data. The unaveraged fit achieves $R^2 = 0.9965$ but tracks the cavity transient rather than spin dynamics. The averaged fit yields $R^2 = 0.246$ with large parameter uncertainties, indicating the Rabi oscillation signal is below the noise floor of the current readout chain.

At $B = 0$ G the same test gives $t \approx 1.0$, $p \approx 0.31$, statistically indistinguishable from zero; the negative numerical sign of the contrast at $B = 0$ G is therefore not a real opposite effect but a fluctuation at the $\sigma \approx 0.35$ mV noise floor. A "difference of differences" test comparing the laser effect across the two field conditions yields $p \approx 0.05$, confirming that the laser-induced modulation is itself B-field-dependent—the signature expected only of spin-cavity coupling, not of laser-induced thermal or stray-light artifacts. At $B = 18$ G, the Zeeman shift tunes the $m_s = 0 \leftrightarrow m_s = +1$ transition into resonance with the cavity at 2.82 GHz, enabling resonant driving with a well-defined transition, a prerequisite for coherent pulsed control.

B. Rabi Dynamics and Cavity Transient Response

Meanwhile, the central finding of the pulsed control experiment is that the τ-sweep data reveals cavity transient dynamics rather than coherent spin control. Without averaging, the reflected amplitude rises monotonically from ≈ 11 mV at $\tau = 20$ ns and saturates at ≈ 17 mV by $\tau = 80$–100 ns, consistent with the cavity ring-up timescale $1/\kappa_c$, not with Rabi oscillations, which would appear as coherent modulation atop this envelope (Fig. 11).

With $N = 512$ averaging, the signal drops to the millivolt scale but the τ-dependent structure remains scattered and statistically inconclusive. Fitting $S(\tau)$ to Eq. (1) yielded

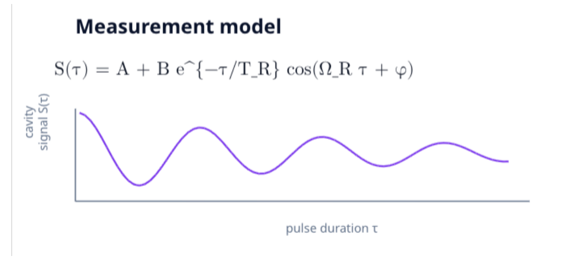

Fig. 11. Expected form of the cavity reflection signal $S(\tau)$ as a function of pulse duration τ. A decaying sinusoid with Rabi frequency Ω_R, decay time T_R, phase ϕ, and offset A is the signature of coherent spin control; deviations from this form, such as the monotonic rise observed in our unaveraged data, indicate the measurement is dominated by cavity transient dynamics rather than spin response.

discrepant parameters between the two datasets: Ω_R ranging from 33 to 70 rad/ns and T_R from 26.74 to 74.72 ns with uncertainties exceeding the central values, indicating the optimizer is fitting noise rather than a physical oscillation. Clean Rabi oscillations remain below the noise floor of the current readout chain.

C. Physical Limitations and Measurement Chain Distortions

The absence of resolved Rabi oscillations is not a failure of the cavity architecture itself, but rather a confluence of physical and experimental limitations distributed across the measurement chain.

We first place our system on the coupling spectrum by estimating the cooperativity

$$C = \frac{4g^2N}{\kappa_c\,\kappa_s} \tag{5}$$

which compares coherent spin–cavity exchange to combined loss rates. Strong coupling ($C \gg 1$) yields resolved avoided crossings and clean vacuum-Rabi oscillations; intermediate coupling ($C \sim 1$) produces a measurable but weak spin-dependent contrast without resolved splitting; weak coupling ($C \ll 1$) leaves the cavity response unperturbed. From the observed cavity ring-up timescale of 80–100 ns we infer a loaded $\kappa_c \approx 2\pi \times 1$–$2$ MHz; with $T_2 \approx 3\,\mu$s typical for room-temperature NV ensembles, $\kappa_s \approx 2\pi \times 100$ kHz. Using the Pre-lab single-spin coupling $g_0 \approx 0.01$ Hz and an NV count $N \sim 10^{14}$ for the diamond volume used here gives $g\sqrt{N} \approx 0.1$–1 MHz, hence C of order 10^{-1}–10^0. This places our system squarely in the intermediate-coupling regime, quantitatively consistent with our absence of resolved avoided crossing in the VNA B-sweep but presence of a statistically significant 13% laser-dependent contrast.

Two physical constraints fundamentally limit coherent control in this architecture. First, the cavity's finite response time ($\kappa_c^{-1} \approx 80$–100 ns) is comparable to or exceeds typical Rabi oscillation periods for moderate drive powers. At pulse durations shorter than this timescale, the intracavity field is still building up, and the effective drive experienced by the spins is severely suppressed relative to the applied rectangular pulse. This validates the distortion mechanism identified in Iriarte-Zendoia *et al.* [7] which showed quantitatively that in cavities with $Q \gtrsim 8 \times 10^3$, the effective drive suppression at

$\tau = 50\,\mathrm{ns}$ is sufficient to reduce the nominal Rabi rotation by a factor of $\sim 3\times$, which at our estimated $\Omega_R \sim$ 10–30 GHz would compress the visible oscillation amplitude below the $\sigma \approx 0.35\,\mathrm{mV}$ noise floor we measure. Second, the collective coupling strength $g\sqrt{N}$ at our diamond and cavity parameters places the system in the intermediate rather than strong coupling regime. In this regime, the spin-cavity interaction is detectable but not so robust as to produce unambiguous dynamical signatures; the Rabi frequency is comparable to the cavity decay rate, and coherent oscillations are damped out on a timescale similar to cavity ring-up. Their result therefore independently predicts our failure to observe clean Rabi oscillations and validates our conclusion that the intermediate coupling regime compounded by pulse distortion, rather than experimental error, is the primary limiting mechanism.

Several instrumentation limitations compound these physical constraints. To begin, our readout oscilloscope has a bandwidth of 300 MHz but must resolve features near the 2.82 GHz carrier frequency. The oscilloscope therefore detects only the *envelope* of the RF signal, discarding high-frequency phase information and introducing aliasing that can smear the true cavity response when the transient timescale is comparable to the oscilloscope's inverse bandwidth. Additionally, the 532 nm laser modulation is controlled via a TTL signal with a bandwidth of approximately 30 kHz, too slow to rapidly establish full optical polarization in rapid-fire sequences. This reduces the effective spin polarization fidelity on each shot, degrading the signal-to-noise ratio of the cavity readout. On the side of the FPGA, DIP-switch-based FPGA pulse gating inherently exhibits finite rise and fall times and may allow microwave leakage during the off-state, introducing CW drive that can depolarize the ensemble or contaminate the averaged waveform. Finally, on the side of the oscilloscope, at short τ values where cavity transients overlap with potential spin-dependent features, manually identifying the dip location using oscilloscope cursors introduces subjective uncertainty, particularly when the signal-to-noise ratio is marginal (Fig. 12).

Of the limitations identified above, the intermediate coupling regime is the most likely cause of the absent Rabi signal. When $g\sqrt{N} < \kappa_c$, coherent spin dynamics are damped faster than they can be resolved, and no amount of averaging or readout improvement can recover an oscillation that is physically washed out Increasing the NV density through a higher-conversion diamond or larger ensemble volume is therefore the most direct path to resolving Rabi oscillations, as it would push the system toward strong coupling where $g\sqrt{N} > \kappa_c$ and coherent modulation becomes observable above the cavity decay rate.

D. Comparison to Prior Work

The literature reports Rabi oscillations in NV ensembles coupled to dielectric cavities [4], but those experiments operated at cryogenic temperatures (T $<$ 10 K) with lower cavity frequencies (200 MHz–1 GHz), reducing both the transient timescale and the dephasing rate, thereby improving the signal-to-noise ratio for coherent oscillations. At room temperature,

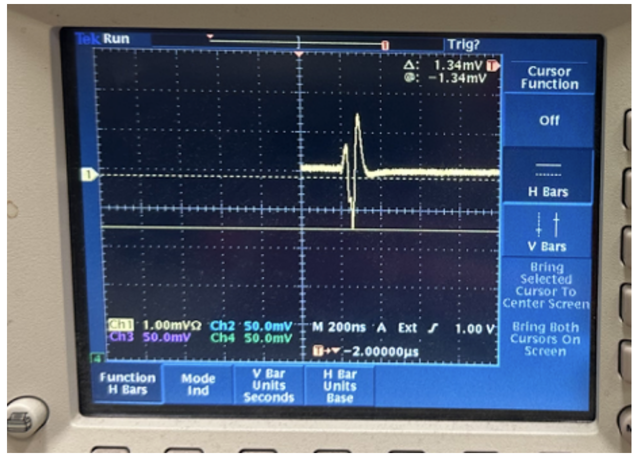

Fig. 12. Oscilloscope trace of the cavity reflection signal at $\tau = 40\,\mathrm{ns}$, showing the characteristic double-transient waveform: a sharp positive spike at pulse onset followed by a negative feature at pulse termination, each corresponding to cavity field build-up and ring-down respectively. The cursor measurement ($\Delta = 1.34\,\mathrm{mV}$) targets the second feature, where the spin-dependent transient is most prominent. The overlap of the two transients at short τ, visible here at 200 ns/div, illustrates the cursor placement ambiguity that introduces subjectivity into the manual amplitude readout.

thermal dephasing and the competing timescales between cavity dynamics and spin precession make observing clean oscillations substantially more challenging.

Wang *et al.* [3] achieved record room-temperature sensitivity using cavity-QED with optical cooling, but their measurement focused on sensitivity rather than time-domain pulse fidelity. Their architecture, while sensitive, does not directly address whether pulsed coherent control can be maintained in high-Q cavities, precisely the question we set out to examine.

The most directly comparable result is Vallabhapurapu et al. [4], who demonstrated Rabi oscillations at Ω_R up to 48 MHz in a $KTaO_3$ dielectric resonator coupled to an NV ensemble at $T < 10\,\mathrm{K}$. Our extracted values, $\Omega_R =$ 33 rad/ns ≈ 5.3 MHz (unaveraged fit, tracking cavity transient) and 70 rad/ns ≈ 11.2 MHz (averaged fit, fitting noise) span a range that straddles but does not converge on a physically consistent value. The cryogenic experiment benefited from both reduced dephasing (longer T_2) and lower cavity frequencies (200 MHz–1 GHz), placing $1/\kappa_c$ well below the Rabi period and avoiding the pulse distortion regime we encounter. Reaching comparable Ω_R at room temperature with our 2.82 GHz, $Q_0 \approx 21{,}363$ cavity would require microwave drive powers roughly 40–60$\times$ higher than currently applied, motivating the power amplifier upgrade recommended in Section VII.A.

E. Implications for Architecture Design

Our results highlight a fundamental trade-off in room-temperature cavity-enhanced NV platforms. **High-Q cavities** maximize readout SNR through strong spin-cavity coupling and frequency selectivity, but their long ring-up times (κ_c^{-1}) distort applied pulses and suppress the effective drive amplitude at short τ, degrading coherent control fidelity. On the other hand, **Moderate-Q cavities** would preserve pulse shape and enable cleaner Rabi observations, but at the cost of reduced readout contrast and higher noise.

The tradeoff suggests that future room-temperature quantum sensing systems should either use pulse-shaping techniques (e.g., GRAPE optimization [7]) to pre-distort the applied drive in a way that compensates for cavity filtering and restores effective rectangular waveforms in the cavity, or employ moderate-Q cavities paired with low-noise amplification, accepting slightly lower direct readout contrast in exchange for preserved pulse fidelity.

VII. Suggestions for Future Development

To resolve coherent Rabi dynamics and accurately extract Ω_R and T_R in this platform, we recommend the following improvements:

A. Hardware

Several hardware changes would most directly improve the platform. First, replacing the 300 MHz oscilloscope with a fast digitizer (GHz-bandwidth ADC) or envelope detector can aid with capturing RF structure with higher fidelity. Optimizing loop-coupler positioning would maximize microwave field homogeneity within the cavity. Increasing microwave drive power, with a power amplifier between the SG384 and circulator if needed, would raise Ω_R above the cavity decay rate, making oscillations faster than decoherence. Most importantly, as we have identified coupling as the most important limiting factor, replacing the current diamond with one of higher NV conversion efficiency would increase $g\sqrt{N}$ and push the system into the strong coupling regime.

B. Measurement

On the measurement side, implementing lock-in detection synchronized to the laser modulation frequency can suppress thermal and RF noise. Increasing averaging depth well beyond N = 512, potentially using on-FPGA accumulation, can avoid oscilloscope memory limits. Replacing manual cursor readout with automated waveform capture and peak-detection would eliminate the subjectivity at short τ and enable denser, unsupervised τ sweeps. A 2D VNA sweep over both τ and drive frequency would also map the full avoided-crossing surface and confirm the operating point more rigorously.

C. FPGA Control

The FPGA itself also has room for extension. A programmable τ register controlled over USB would eliminate the need to reflash the bitstream between points, enabling automated dense sweeps from Python. A software-controlled trigger that interleaves laser-on and laser-off cycles within the same pulse sequence would eliminate drift between the two contrast conditions. Finally, building averaging accumulators directly on the FPGA will allow us to compute trace averages without relying on scope memory.

VIII. Conclusions

In this project, we have designed and implemented an FPGA-based pulsed control platform for cavity-enhanced room-temperature NV magnetometry, extending the existing steady-state frequency-domain architecture into the time domain. End-to-end pulsed operation of the full measurement chain was demonstrated, and a 13% spin-dependent contrast at $B = 18\,\mathrm{G}$, confirmed as genuine spin-cavity coupling through a $B = 0\,\mathrm{G}$ control, validated both the cavity-enhanced readout architecture and the ensemble approach, where collective coupling $g\sqrt{N}$ enables room-temperature detection without cryogenic amplification.

Clean Rabi oscillations remain unresolved in the τ-sweep data (Fig. 9), buried beneath a combination of cavity transient dynamics, measurement noise, and spin dephasing. Coherent modulation could not be reliably extracted from the signal envelope. Curve-fitting to the standard Rabi model (Eq. 1) yielded unreliable parameter extraction (Fig. 10) with large, overlapping error bars, precluding accurate determination of Ω_R and T_R. As discussed, the most fundamental limiting factor is the intermediate spin-cavity coupling regime. Increasing the NV ensemble density is therefore the highest-priority path forward, alongside the hardware and FPGA extensions suggested in the previous section.

More broadly, despite not yet resolving clean Rabi oscillations, this work takes an essential step toward compact, room-temperature quantum sensing architectures. We have validated that cavity-enhanced NV platforms can support pulsed readout with statistically significant spin-dependent contrast, and we have identified the specific physical and experimental bottlenecks limiting coherent control, which is information vital for informing the design of next-generation systems. The tradeoff between cavity Q and pulse fidelity that we document here may prove relevant not only for NV ensembles but also for other solid-state quantum platforms (e.g., rare-earth ions, silicon carbide defects) exploiting cavity enhancement for room-temperature sensing.

Such sensors promise transformative applications in neural magnetic imaging, materials characterization, and navigation, but only if both readout fidelity and coherent control can be simultaneously achieved. Our results suggest that achieving both likely requires either pulse shaping, intermediate-Q cavity designs, or hybrid architectures that trade some readout sensitivity for control fidelity, a design principle that should inform the community's approach to cavity-enhanced quantum metrology in the coming years.

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